THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** $(-5, -2)$	Reflection in the y-axis changes the sign of the abscissa.
2	**(a)** $3x + 4$	Linear means degree exactly 1.
3	**(c)** 3	It is the larger of the two powers.
4	**(b)** $4x^2 + 12xy + 9y^2$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(b)** 48	Chord $= 2\sqrt{r^2 - d^2}$.
6	**(c)** 84	Use Heron's formula.
7	**(b)** $\frac{11}{50}$	Relative frequency of the outcome.
8	**(b)** 53	$2 \times 25 + 3$.
9	**(b)** If a number is divisible by 3, then it is divisible by 6.	Interchange the hypothesis and the conclusion.
10	**(a)** 44	$\frac{27 \times 57 + 39 \times 35}{66} = 44$.
11	**(d)** 2	Carries out of the columns: [1, 0, 1, 0].
12	**(d)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
13	**(a)** No	$1(-3) + (2)(2) = 1$, not equal to 2.
14	**(b)** 242 cm^2	$2\pi rh$ with $r = 3.5$, $h = 11$.
15	**(d)** 5	Apply the distance formula.
16	**(a)** -10	Put x in $p(x) = 0$.
17	**(a)** $\frac{3}{11}$	Multiply by 100 and subtract.
18	**(c)** $9x^2 - 12xy + 4y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
19	**(a)** 15	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
20	**(a)** 21	$s = \frac{a+b+c}{2}$.
21	**(a)** $\frac{7}{25}$	Relative frequency.
22	**(a)** 17th term	$3n + 2 = 53$ gives $n = 17$.
23	**(d)** False	Counterexample: 9 is divisible by 3 but not by 6.
24	**(a)** 33	$\frac{14 \times 31 + 61}{15} = 33$.
25	**(c)** 6	Divisors of 44: 1, 2, 4, 11, 22, 44.
26	**(a)** 7	Opposite angles are equal.
27	**(c)** 4	Put x=0: $y = 4$.
28	**(c)** 671 cm^2	$2\pi r(r + h)$ with $r = 3.5$, $h = 27$.
29	**(d)** $(2, 2)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
30	**(b)** -1	Substitute $x = -k$.
31	**(c)** $\sqrt{5}$	$2^2 < 5 < 9$
32	**(c)** $x^2 + 4x - 21$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
33	**(a)** 37°	Half of the central angle.
34	**(b)** 616	Find r from $C = 2\pi r$, then use $A = \pi r^2$.
35	**(d)** $\frac{5}{9}$	Count vowels among all the letters (repeats counted).
36	**(a)** 19	AP with $a = 4$, $d = 3$: $t_6 = 4 + 5 \times 3$.
37	**(c)** $a = -3,\ b = 2$	$a^2 = 9 > b^2 = 4$.
38	**(b)** 69	$\frac{70 \times 6 + 65 \times 6 + 75 \times 3}{15} = 69$.

39	**(d)** 2	Pairs: $(90,82)\to(82,8)\to(74,8)\to(66,8)\to(58,8)\to(50,8)\to(42,8)\to(34,8)\to(26,8)\to(18,8)\to(10,8)\to(8,2)\to(6,2)\to(4,2)\to(2,2)\to(2,0)$.
40	**(a)** 20°	Adjacent angles add up to 180°.
41	**(b)** -9	$\dfrac{-2-(7)}{5-(4)}=-9$.
42	**(c)** 2200 cm^2	πrl with the given values.
43	**(b)** $(6,0)$	Ordinate is 0 on the x-axis.
44	**(c)** 7	Put $x=2$.
45	**(c)** irrational	Use the rules for sums and products of rational and irrational numbers.
46	**(a)** 10609	Write $103=100+3$
47	**(a)** 147°	Angles on the two arcs are supplementary: $180^\circ-t$.
48	**(c)** 176	Distance in one turn $=$ circumference.
49	**(b)** $\frac{3}{8}$	Use the 8 equally likely outcomes.
50	**(c)** 47	$a+(n-1)d=5+14\times(3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 9	List the factors of 36 in pairs; the count is 9 (odd, so 36 is a perfect square).
2	**(b)** 27%	$\frac{60\times15+120\times33}{180}=27\%$.
3	**(c)** 2	$146=3\times48+2$.
4	**(a)** 28 cm	Diagonals bisect each other, so $AC=2\times OA$.
5	**(d)** 1	Rewrite as $y=x+\ldots$; slope $=-\frac{a}{b}$.
6	**(b)** 1617 cm^3	$\frac{1}{3}\pi r^2h$ with the given values.
7	**(c)** Quadrant II	Use the sign rules for the quadrants.
8	**(a)** $(4,0)$	Put $y=0$.
9	**(c)** $\frac{1}{6}$	Use the repeating-decimal method.
10	**(a)** $(x+4)$	$a^2+2ab+b^2=(a+b)^2$
11	**(d)** 55°	The exterior angle equals the interior opposite angle.
12	**(b)** 70	Area $=\frac{1}{2}d_1d_2$.
13	**(a)** $\frac{3}{4}$	Count the balls that are not black.
14	**(b)** 210	$S_n=\frac{n}{2}[2a+(n-1)d]$ with $a=3,d=4$.
15	**(d)** right-angled	$9^2+40^2=1681$ and $41^2=1681$.
16	**(c)** 153	17% of $900=\frac{17}{100}\times900=153$.
17	**(a)** $1,2,\ldots,n$	Every j from 1 to n is tested.
18	**(d)** parallelogram	A test for a parallelogram.
19	**(a)** 8	$k(5)+(2)=42$ gives $k=8$.
20	**(d)** 15400 cm^2	$4\pi r^2$ with $r=35$.
21	**(d)** 7	The distance from the x-axis is $\lvert y\rvert$.
22	**(c)** 7	Find each zero.
23	**(c)** 8	Use Brahmagupta's rules.
24	**(b)** $(x+2)(x+5)$	Find two numbers with the given sum and product.
25	**(c)** 50°	$OA=OB$ (radii) so the base angles are equal.
26	**(b)** 36	Use Brahmagupta's formula $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
27	**(c)** 70	$P(\text{not } E)=1-P(E)$.
28	**(a)** 162	$ar^{n-1}=2\times3^4$.
29	**(d)** 10	$n=10$ gives 121, which is composite.
30	**(a)** $\frac{w_1x_1+w_2x_2}{w_1+w_2}$	Formula.
31	**(d)** 6	Pairs: $(90,50)\to(50,40)\to(40,10)\to(30,10)\to(20,10)\to(10,10)\to(10,0)$.
32	**(a)** 25°	A diagonal of a rhombus bisects the angle at the vertex.
33	**(b)** $(6,-4)$	$8(6)+9(-4)=12$
34	**(b)** 130977 cm^3	$\frac{4}{3}\pi r^3$ with $r=31.5$.
35	**(a)** $(8,11)$	$B=2M-A$ coordinatewise.
36	**(b)** $6n-5$	Use $a_n=a_1+(n-1)d$.
37	**(a)** $\frac{7}{16}$	Terminates when the denominator has only 2 and 5 as prime factors.
38	**(c)** 53	$x^2+y^2=(x+y)^2-2xy$.
39	**(d)** equidistant	Recall the theorem.
40	**(b)** 44	Arc $=\frac{\theta}{360}\times2\pi r$.

41	**(c)** $\frac{1}{3}$	Count favourable outcomes out of 6.
42	**(a)** 21	$3 + (n-1)4 = 83$ gives $n = 21$.
43	**(d)** a counterexample	One counterexample is enough.
44	**(c)** 30	Sum of the extra numbers $= 46 \times 45 - 30 \times 54 = 450$; divide by 15.
45	**(b)** 449	Apply the rule each time.
46	**(c)** 43 cm	Each new side is half of a side of the original triangle.
47	**(a)** -3	$y = mx + c$ with $y = -2$ at $x = -1$: $c = -2 - (-1)(-1)$.
48	**(a)** 77 cm^2	$2\pi r^2$ with $r = 3.5$.
49	**(a)** $(3, 8)$	Reflection in the y-axis changes the sign of the abscissa.
50	**(c)** 250	Fare $=$ fixed $+$ rate $\times$ time.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	It is the larger of the two powers.
2	**(b)** $a^3 + 3a^2b + 3ab^2 + b^3$	Standard cube identity.
3	**(a)** 14	Chord $= 2\sqrt{r^2 - d^2}$.
4	**(d)** 22	Perimeters are equal.
5	**(a)** $0 \le P(E) \le 1$	Recall the definition.
6	**(b)** an arithmetic progression	Definition of an AP.
7	**(c)** 25	Only a perfect square has an odd number of factors; $25 = 5^2$.
8	**(d)** 31	$\dfrac{8 \times 32 - 39}{7} = 31.$
9	**(b)** 3	Pairs: $(138, 117) \to (117, 21) \to (96, 21) \to (75, 21) \to (54, 21) \to (33, 21) \to (21, 12) \to (12, 9) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$.
10	**(c)** 32 cm^2	Area $= \frac{1}{2}d^2$.
11	**(d)** 0	A horizontal line has slope 0; a vertical line has no (undefined) slope.
12	**(a)** $\frac{1}{3}Bh$	One third of the prism.
13	**(b)** 13	Apply the distance formula.
14	**(d)** $5 - 2x$	Linear means degree exactly 1.
15	**(c)** $\frac{5}{11}$	Multiply by 100 and subtract.
16	**(d)** $9x^2 + 24xy + 16y^2$	$(a + b)^2 = a^2 + 2ab + b^2$
17	**(c)** 20	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
18	**(b)** 42	Use Heron's formula.
19	**(c)** $\frac{7}{50}$	Relative frequency of the outcome.
20	**(b)** 44	$3 \times 16 - 4$.
21	**(b)** If a quadrilateral is a rectangle, then it is a square.	Interchange the hypothesis and the conclusion.
22	**(a)** 54	$\dfrac{35 \times 65 + 55 \times 47}{90} = 54.$
23	**(d)** 1	Carries out of the columns: [0, 1, 0, 0].
24	**(c)** 180°	One part $= 360^\circ \div 12 = 30^\circ$.
25	**(d)** Yes	$6(5) + (4)(-6) = 6$, equal to 6.
26	**(c)** 4752 cm^2	$2\pi rh$ with $r = 28$, $h = 27$.
27	**(a)** $(-2, 4)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
28	**(d)** -12	Put x in $p(x) = 0$.
29	**(d)** $\sqrt{11}$	$3^2 < 11 < 16$
30	**(d)** $4x^2 - 20xy + 25y^2$	$(a - b)^2 = a^2 - 2ab + b^2$
31	**(b)** 44°	Half of the central angle.
32	**(d)** 18	$s = \frac{a+b+c}{2}$.
33	**(b)** $\frac{9}{40}$	Relative frequency.
34	**(c)** 21th term	$4n - 3 = 81$ gives $n = 21$.
35	**(b)** False	Counterexample: a 4 cm by 6 cm rectangle is not a square.
36	**(b)** 32	$\dfrac{21 \times 31 + 53}{22} = 32.$
37	**(a)** 4	Divisors of 77: 1, 7, 11, 77.
38	**(a)** 29	Opposite angles are equal.
39	**(b)** $\frac{59}{5}$	Put y=0: $x = \frac{59}{5}$.

40	**(c)** 1881 cm^2	$2\pi r(r+h)$ with $r = 10.5$, $h = 18$.
41	**(b)** $(-4, 0)$	Ordinate is 0 on the x-axis.
42	**(b)** -1	Substitute $x = -k$.
43	**(b)** irrational	Use the rules for sums and products of rational and irrational numbers.
44	**(b)** $x^2 + 3x - 40$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
45	**(c)** 131°	Angles on the two arcs are supplementary: $180^\circ - t$.
46	**(b)** 1386	Find r from $C = 2\pi r$, then use $A = \pi r^2$.
47	**(d)** $\frac{1}{3}$	Count vowels among all the letters (repeats counted).
48	**(b)** 29	AP with $a = 9$, $d = 4$: $t_6 = 9 + 5 \times 4$.
49	**(a)** $n = 40$	$1600 + 40 + 41 = 1681 = 41^2$.
50	**(b)** 71	$\dfrac{93 \times 3 + 80 \times 2 + 57 \times 6}{11} = 71$.

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